

Sequences and series of real numbers

1. Sequences of real numbers

Def. 1) $f: \mathbb{N}^* \rightarrow \mathbb{R}$ is called seq. of real numbers

Denote: $a_n = f(n)$, $n \geq 1$, $(a_n)_{n \geq 1}$

Def. 2) A seq. a_n is called increasing or decreasing if $a_n \leq a_{n+1}$, $(a_n \geq a_{n+1})$, $\forall n \geq 1$

$a_1 \leq a_2 \leq \dots \leq a_n$ - increasing

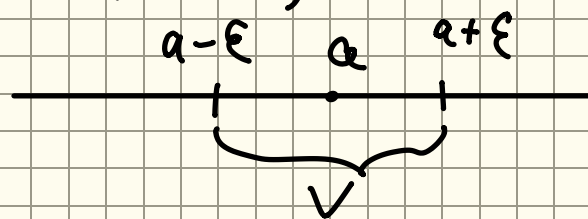
$a_1 \geq a_2 \geq \dots \geq a_n$ - decreasing

Def. 3) a_n is called upper (lower) bounded if $\exists M \in \mathbb{R} (\exists m \in \mathbb{R})$ if $a_n \leq M$ ($a_n \geq m$), $\forall n \geq 1$. If $m \leq a_n \leq M$, $\forall n \geq 1$, then a_n is called a bounded sequence.

Supremum a_n , $\forall n \geq 1 = M$

Infimum a_n , $\forall n \geq 1 = m$

Def. 4) The $V \subseteq \mathbb{R}$ is called neighbourhood of $a \in \mathbb{R}$ if $\exists (a - \varepsilon, a + \varepsilon) \subseteq V$, $\varepsilon > 0$.

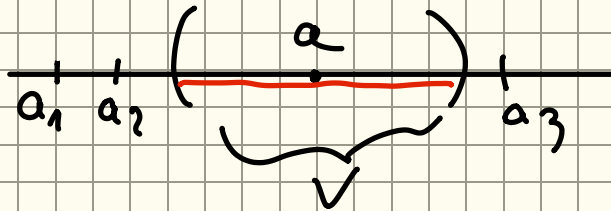


Denote: by $\mathcal{V}(a)$ the set of all neighbourhoods of a .

Denote: $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$

Def. 5) $V \subseteq \mathbb{R}$ is called neigh. of $+\infty$ ($-\infty$) if $\exists (a, +\infty) \subseteq V$ ($\exists (-\infty, a) \subseteq V$)

Def. 6) (Limit) An element $a \in \overline{\mathbb{R}}$ is called limit of $(a_n), n \geq 1$ if $\forall V \in \mathcal{V}(a) \exists n_V \in \mathbb{N}^*$ s.t. $a_n \in V, \forall n \geq n_V$.



Denote: $\lim_{n \rightarrow \infty} a_n$ or $\lim a_n$ or $a_n \rightarrow a$

Def. 7) A sequence having a finite limit is called convergent.

Th. 8) $(a_n), n \geq 1$ is conv. $\Leftrightarrow \forall \varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}$ s.t. $|a_n - a| < \varepsilon, \forall n \geq n_\varepsilon$.

Th. 9) Every monotonic and bounded seq. is convergent (Weierstrass).

Remark: A conv. seq. is bounded but not necessarily monotonic.

$$\text{ex.: } a_n = \frac{(-1)^n}{n} \rightarrow 0$$

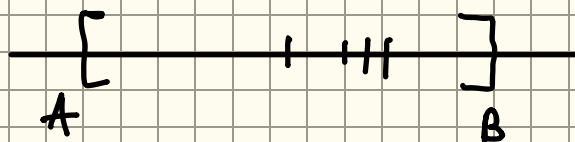
Def. 10) (Subsequence) Let $(a_n), n \geq 1$ and $n_1 < n_2 < n_3 < \dots$ a strictly increasing seq. of natural numbers. Then $a_{n_1}, a_{n_2}, \dots$ denoted by $(a_{n_k}), k \geq 1$ is called subsequence.

ex.: $a_1, a_3, a_5, \dots$
 $a_1, a_4, a_{16}, \dots$

Properties:

- If a seq. has a limit, every subseq. has the same limit.
- Every seq. has a monotonic subseq.

Th. 11) (Cesaro Lemma) Every bounded seq. contains at least one conv. subseq. (Bolzano-Weierstrass th.).



Def. 12) (Fundamental seq.) $(a_n), n \geq 1$ is called fundamental seq. if $\forall \epsilon > 0, \exists n_\epsilon \in \mathbb{N}^*$ s.t. $\forall n \geq n_\epsilon, \forall p \in \mathbb{N}: |a_{n+p} - a_n| < \epsilon$ (Cauchy seq.)

Th. 13) A seq. of real numbers is conv. $\Leftrightarrow$ it is a fundamental seq.

Application: Study the conv. of $(a_n), n \geq 1$

$$a_n = \frac{n \cdot n!}{1^2} + \frac{n \cdot n^2}{2^2} + \dots + \frac{n \cdot n^n}{n^2}, n \geq 1$$

$$|a_{n+p} - a_n| \xrightarrow{n \rightarrow \infty} 0, \forall p \in \mathbb{N}$$

$$|a_{n+p} - a_n| = \left| \frac{n \cdot n \cdot (n+1)}{(n+1)^2} + \dots + \frac{n \cdot n \cdot (n+p)}{(n+p)^2} \right| \leq \left| \frac{n \cdot n \cdot (n+1)}{(n+1)^2} \right| + \dots$$

$$\left| \frac{a_{m+p} - a_m}{(m+p)^2} \right| \leq \frac{1}{(m+1)^2} + \dots + \frac{1}{(m+p)^2} \leq \frac{1}{n(m+1)} + \frac{1}{(m+1)(m+2)} + \dots + \frac{1}{(m+p-1)(m+p)}$$

$$= \frac{1}{n} - \frac{1}{n+1} + \frac{1}{n+1} - \frac{1}{n+2} + \dots + \frac{1}{n+p-1} - \frac{1}{n+p} = \frac{1}{n} - \frac{1}{n+p} < \frac{1}{n}$$

$$|a_{m+p} - a_m| < \frac{1}{n}, \forall n, p \in \mathbb{N}^* \Rightarrow |a_{m+p} - a_m| \xrightarrow{n \rightarrow \infty} 0$$

$$\Rightarrow (a_n) \text{ is fundamental} \Rightarrow (a_n) \text{ is convergent.}$$

2. Upper and Lower Limit

Def. 1) $a \in \overline{\mathbb{R}}$ is called limit point (cluster point or accumulation point) of $(a_n), n \geq 1$ if $\exists (a_{n_k}), k \geq 1$ s.t. $a_{n_k} \xrightarrow{k \rightarrow \infty} a$

Denote: $L(a_n)$ the set of all limit points of $(a_n), n \geq 1$.

Def. 2) $\limsup a_n = \sup L(a_n), n \in \mathbb{N}^*$
 $\liminf a_n = \inf L(a_n), n \in \mathbb{N}^*$
 ($\overline{\lim} a_n$ resp. $\underline{\lim} a_n$)

$$\text{ex.: } a_n = (-1)^{n-1}, n \geq 1$$

$$\begin{array}{l} 1, -1, 1, -1, \dots \\ a_{2n} = -1 \rightarrow -1 \\ a_{2n-1} = 1 \rightarrow 1 \end{array} \left| \Rightarrow L(a_n) = \{-1, 1\} \Rightarrow \right.$$

$$\limsup a_n = 1$$

$$\liminf a_n = -1$$

2) Consider the sequence:

$$1, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{3}, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

$$L(a_n) = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\} \cup \{0\}$$

$$\liminf a_n = 0$$

$$\limsup a_n = 1$$

$$3) (a_n)_{n \geq 1}, a_n = \sin n$$

$$L(a_n) = [-1, 1]$$

$$\limsup a_n = 1$$

$$\liminf a_n = -1$$

Th. 3) $(a_n)_{n \geq 1}$ has limit $\Leftrightarrow \limsup a_n = \liminf a_n$.

3. Series of numbers

$(a_n)_{n \geq 1}$ is a seq. and consider

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots$$

$$\text{ex.: 1) } \sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1}{2} + \frac{2}{2^2} + \dots$$

$$2) \sum_{n=1}^{\infty} (-1)^{n-1} = 1 - 1 + 1 - 1 + \dots$$

$$(1-1) + (1-1) + \dots = 0 + 0 + 0 + \dots = 0$$

$$1 + (1-1) + (1-1) + \dots = 1 + 0 + 0 + \dots = 1$$

Obs. Properties of finite sums used for infinite sums lead to contradictions.

Def. 1) Let $\sum_{n=1}^{\infty} a_n$ be a series. a_n is called

general term. Consider $(r_n), n \geq 1$:

$r_n = a_1 + a_2 + \dots + a_n, n \geq 1$ (seq. of partial sums)

Def. 2) If $\exists \lim r_n = r, r \in \mathbb{R}$, then r is called sum of the series.

Denote: $\sum_{n=1}^{\infty} a_n = r$

Obs. If r is finite then the series is called convergent.

ex.: 1) Geometric series

$$\sum_{n=0}^{\infty} q^n = 1 + q + q^2 + \dots, q \in \mathbb{R} \text{ is convergent}$$

$$\Leftrightarrow |q| < 1 \Rightarrow \sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$$

Proof: $r_n = \sum_{n=0}^{\infty} q^n = \begin{cases} \frac{1-q^{n+1}}{1-q}, & q \neq 1 \\ n+1, & q = 1 \end{cases}$

2) Harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \dots = \infty$$

3) Generalized harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}} = \frac{1}{1^{\alpha}} + \frac{1}{2^{\alpha}} + \dots, \alpha \in \mathbb{R} \text{ is convergent}$$

$$\Leftrightarrow \alpha > 1$$

Definition: $\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}} = \zeta(\alpha)$, ζ - Zeta function,
 $\alpha > 1$

Particular cases:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \text{ (Euler)}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$